Wilson Lattice Gauge Theory on the GeoVac Label Graphs: Exact Factorization on the Scalar Graph and Compact $U(1)$ on the Dirac Rule-B Graph

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Abstract

The GeoVac label graphs carry Wilson lattice gauge theories, and the two graphs studied here sit at opposite ends of solvability. On the scalar label graph, whose vertices are the labels (n, l, m) of the S^3 harmonics, no edge changes l, and each l-block is a planar $(n_{\max} - l) \times (2l + 1)$ grid. Taking the plaquettes to be the graph's 4-cycles (the unit squares of these grids), the plaquettes form a basis of the cycle space at every $n_{\max}$, so after gauge fixing on a spanning tree the plaquette holonomies are independent Haar variables. SU(2) Wilson theory on this graph is therefore the exactly solvable two-dimensional lattice gauge theory: $Z(\beta) = [2e^{-\beta} I_1(\beta)/\beta]^{\beta_1}$, and every closed loop enclosing A plaquettes has $\langle W \rangle = (I_2(\beta)/I_1(\beta))^A$, an area law by factorization that carries no information about confinement. The weak-coupling expansion gives the curl form $(\beta/8) \sum_a \langle A^a, B_2 B_2^\top A^a \rangle$, which is supported on the cycle space and completes the edge Laplacian $B^\top B$ to the full Hodge-1 Laplacian of the plaquette-filled complex; the cubic term is nonzero, with a vertex proportional to ϵ^{abc}. Restricting the links to the maximal torus gives compact $U(1)$ Wilson theory, with $\langle W \rangle = (I_1/I_0)^A$, and we record the abelian dictionary between Wilson and graph-Hodge data together with the exact $n_{\max} = 3$ edge-Laplacian spectrum. On the Dirac Rule-B graph, whose edges are E1 dipole transitions and all change l, the graph is connected, its 4-cycles outnumber the independent cycles (44 against 11 at $n_{\max} = 2$) and close into 2-cycles of three plaquettes, so compact $U(1)$ does not factorize. Seven standard diagnostics are compatible with compact $U(1)$ and discriminate no dimension: the monopole density falls roughly exponentially in β on a range below the four-dimensional transition, full Monte Carlo Wilson loops resolve no area law (the area coefficient of a joint area-plus-perimeter fit is consistent with zero at $n_{\max} = 2$), and the Polyakov loop on the product graph with a time circle is well defined through one extra cohomology class, while its expectation vanishes by centre symmetry. Each Rule-B link lies in many plaquettes (8.8 and 37.5 on average at $n_{\max} = 2, 3$), so the comparison class is mean-field or large-dimension lattice gauge theory, not a three-dimensional lattice. The label graphs have no continuum limit, and nothing here bears on one.

1 Introduction

The GeoVac framework works on finite graphs whose vertices are quantum numbers. The scalar label graph has the labels (n, l, m) of the unit-three-sphere harmonics as vertices [21]; its edge Laplacian preserves l, splits into $n_{\max}$ disconnected blocks, and does not converge to the S^3 Laplacian (`tests/test_graph_s3_nonconvergence.py`). The Dirac Rule-B graph of Paper 29 [23] has the Dirac labels (n, κ, m_j) as vertices and the E1 dipole transitions as edges. Any oriented graph supports a Wilson lattice gauge theory [1, 2, 3] once a set of plaquettes is chosen, and this paper works out what that theory is on these two graphs.

The two answers are opposite. On the scalar graph each l-block is a planar grid, its unit squares are its 4-cycles, and they form a basis of the cycle space. The theory is then the exactly solvable two-dimensional lattice gauge theory [5, 7, 8]: plaquette holonomies are independent, the partition function and every Wilson loop are products of one-plaquette integrals, and the area law holds by factorization at every coupling. We state this for SU(2) (Section 3), give the weak-coupling kinetic term and its place in the discrete Hodge decomposition (Section 3.4), and take the abelian section by restricting to the maximal torus (Section 4). On the Rule-B graph the 4-cycles are far more numerous than the independent cycles and satisfy relations, so compact $U(1)$ is a genuinely interacting theory there; Section 5 reports seven standard diagnostics, all of which it passes in the sense of being compatible with compact $U(1)$, and none of which discriminates a dimension.

Scope. The constructions are finite, Euclidean, and Haar-normalizable; no renormalization enters. This paper does not address a Yang–Mills mass gap, a continuum limit (the label graphs do not converge to a manifold), or the fine-structure constant. The results on the scalar graph are the textbook two-dimensional results applied to a disjoint union of planar grids; what the paper adds is the identification of the label graphs' plaquette complexes and the measured Rule-B diagnostics. The construction is an instance of the gauge-network framework of Marcolli and van Suijlekom [16], whose spectral action on a graph is a Wilson lattice gauge theory; its continuum limit is Yang–Mills without a Higgs field [17]. For SU(2) Hamiltonian lattice gauge theory built for quantum simulation from discrete partitionings of the group, see Refs. [18, 19].

2 The label graphs and their plaquette complexes

2.1 The scalar label graph

The scalar graph at cutoff $n_{\max}$ has vertices (n, l, m) with $1 \le n \le n_{\max}$, $0 \le l \le n-1$, $|m| \le l$, so $|V| = \sum_n n^2$. Its edges are the angular ladders $(n, l, m) \leftrightarrow (n, l, m \pm 1)$ and the radial ladders $(n, l, m) \leftrightarrow (n \pm 1, l, m)$. No edge changes l, so the graph is the disjoint union

$$G_{\text{scalar}} \;=\; \bigsqcup_{l=0}^{n_{\max}-1} P_{n_{\max}-l} \times P_{2l+1}, \tag{1}$$

where P_k is the path on k vertices and $\times$ is the Cartesian product: the l-block is a planar grid with rows indexed by $n = l+1, \dots, n_{\max}$ and columns by m. A $k_1 \times k_2$ grid has $(k_1-1)(k_2-1)$ unit squares, so

$$\beta_1 \;=\; |E| - |V| + c \;=\; \sum_{l=0}^{n_{\max}-1} 2l\,(n_{\max} - l - 1), \tag{2}$$

with $c = n_{\max}$ components (Table 1).

At $n_{\max} = 3$ the only cycles are the two unit squares of the p-block,

$$\begin{aligned} P_A &: (2,1,-1) \to (2,1,0) \to (3,1,0) \to (3,1,-1) \to (2,1,-1), \\ P_B &: (2,1,0) \to (2,1,1) \to (3,1,1) \to (3,1,0) \to (2,1,0), \end{aligned} \tag{3}$$

which share the edge $\{(2,1,0), (3,1,0)\}$.

Table 1: The scalar label graph. c is the number of components, $\beta_1 = E - V + c$ the first Betti number, and N_4 the number of (unoriented) 4-cycles, which equals β_1 at every cutoff.

$n_{\max}$	V	E	c	β_1	N_4
2	5	3	2	0	0
3	14	13	3	2	2
4	30	34	4	8	8
5	55	70	5	20	20

2.2 Plaquettes and the Hodge operators

Definition 1 (Plaquettes)**.** *The plaquette set $\mathcal{P}$ of a label graph is its set of 4-cycles, each taken once with a fixed orientation.*

On the scalar graph the 4-cycles are exactly the unit squares of the grids, and $N_4 = \beta_1$ (Table 1). Longer closed walks add nothing to the plaquette set: at $n_{\max} = 4$ there are 7 primitive non-backtracking closed walks of length 6 and 18 of length 8 (one per unoriented class), and the signed edge vector of every closed walk lies in the cycle space, which the 4-cycles span.

Fix an orientation on each edge. The signed vertex–edge incidence matrix $B \in \mathbb{Z}^{V\times E}$ has $B_{v,e} = \pm 1$ at the two ends of e, and the signed edge–plaquette incidence matrix $B_2 \in \mathbb{Z}^{E\times\mathcal{P}}$ has $(B_2)_{e,P} = \pm 1$ according to whether P traverses e along or against its orientation. As maps on cochains, $B^\top$ is the coboundary from vertex functions to edge functions and $B_2^\top$ is the coboundary from edge functions to plaquette functions: $(B_2^\top a)_P = \sum_{e\in P} \pm a_e$ is the circulation of a around P. Since every plaquette is closed, $BB_2 = 0$. The combinatorial Hodge Laplacians [14, 15] are

$$L_0 = BB^\top, \qquad L_1 = B^\top B, \qquad \Delta_1 = B^\top B + B_2 B_2^\top. \tag{4}$$

$L_0 = D - A$ is the graph Laplacian; L_1 is the edge Laplacian, whose kernel is the cycle space $\ker B$, of dimension β_1; and Δ_1 is the Hodge-1 Laplacian of the 2-complex obtained by attaching the plaquettes as faces. The nonzero spectra of L_0 and L_1 coincide, being the squared singular values of B. The kernel of Δ_1 is the first cohomology of the 2-complex,

$$\dim\ker\Delta_1 \;=\; \beta_1 - \operatorname{rank} B_2, \tag{5}$$

which vanishes exactly when the plaquettes span the cycle space. When in addition $\operatorname{rank} B_2 < |\mathcal{P}|$, the plaquettes satisfy relations; the kernel of B_2, of dimension $b_2 = |\mathcal{P}| - \operatorname{rank} B_2$, consists of the closed 2-cycles of the complex.

On the scalar graph B_2 has full rank $|\mathcal{P}| = \beta_1$: the plaquettes are a basis of the cycle space and $b_2 = 0$ (`tests/test_su2_wilson_gauge.py`, class `TestCertRunCoverageClosures`, `test_plaquettes_form_cycle_basis`, checked at $n_{\max} = 3, 4, 5$). Every edge of a path-like block ($l = 0$, and the $l = n_{\max} - 1$ block) is a bridge, lying in no plaquette.

2.3 The Dirac Rule-B graph

The Rule-B graph [23] has vertices (n, κ, m_j) with $1 \le n \le n_{\max}$, $\kappa \in \mathbb{Z}\setminus\{0\}$ encoding $l = -\kappa - 1$ ($\kappa < 0$) or $l = \kappa$ ($\kappa > 0$) and $j = |\kappa| - \frac{1}{2}$, and $|m_j| \le j$; there are $2n^2$ vertices in shell n. Two vertices are adjacent when

$$|\Delta n| \le 1, \quad |\Delta l| = 1, \quad |\Delta j| \le 1, \quad |\Delta m_j| \le 1, \tag{6}$$

Table 2: The Dirac Rule-B graph. $\langle q_e \rangle$ is the mean number of 4-cycles through an edge; on $\mathbb{Z}^d$ it is $2(d-1)$.

$n_{\max}$	V	E	c	β_1	N_4	$\langle q_e \rangle$
2	10	20	1	11	44	8.80
3	28	106	1	79	994	37.51
4	60	312	1	253	5,047	–
5	110	692	1	583	14,881	–

the E1 dipole selection rule. Every edge changes l by one, so the graph is bipartite under $(-1)^l$ and contains no triangles. It is connected at every cutoff tested (Table 2), with mean degree $2E/V = 4.0$, 7.6, 10.4, 12.6 at $n_{\max} = 2$–5. Its 4-cycles outnumber its independent cycles by a factor that grows with the cutoff, and at $n_{\max} = 2$, where the scalar graph has no cycles at all, Rule B already has 44 4-cycles. The rank of B_2 equals $\beta_1 = 11$ at $n_{\max} = 2$, so the plaquettes span the cycle space ($\dim\ker\Delta_1 = 0$) and satisfy $b_2 = 44 - 11 = 33$ relations. The smallest closed 2-cycle consists of three plaquettes, each pair sharing two edges, with the six edges each used twice; there are 60 such 2-cycles at $n_{\max} = 2$ and 80 at $n_{\max} = 3$. On the hypercubic lattice $\mathbb{Z}^d$, $d \geq 3$, the smallest closed 2-surface is the six-plaquette boundary of a cube.

3 SU(2) Wilson theory on the scalar graph

3.1 Link variables, action, gauge invariance

Each oriented edge $u \to v$ carries $U_{u\to v} \in \mathrm{SU}(2)$, with $U_{v\to u} = U_{u\to v}^\dagger$. For a plaquette traversed as $e_1e_2e_3e_4$ the holonomy is $U_P = U_{e_1}U_{e_2}U_{e_3}U_{e_4}$, and the Wilson action is

$$S_W[U] = \beta \sum_{P\in\mathcal{P}} \Big(1 - \tfrac{1}{2}\mathrm{Re}\,\mathrm{tr}\,U_P\Big), \qquad \beta = 4/g^2. \tag{7}$$

Under $U_{u\to v} \mapsto g_u U_{u\to v} g_v^\dagger$ with $g : V \to \mathrm{SU}(2)$ each holonomy is conjugated, so S_W is gauge invariant; with random Haar links and gauge transformations at $n_{\max} = 3$ the change in S_W is at machine precision (`tests/test_su2_wilson_gauge.py`, class `TestGaugeInvariance` and `test_s3_coulomb_max_n3_endtoend`, which assert $< 10^{-13}$). The partition function is $Z(\beta) = \int \prod_e dU_e\, e^{-S_W[U]}$ with normalized Haar measure.

3.2 Character expansion

With $\chi_j(U) = \sin((j+1)\theta)/\sin\theta$ the character of the $(j+1)$-dimensional irreducible representation at $U = e^{i\theta\hat{n}\cdot\boldsymbol{\sigma}}$, the one-plaquette Boltzmann factor expands as [4, 2]

$$\exp\Big(\tfrac{\beta}{2}\mathrm{Re}\,\mathrm{tr}\,U\Big) = \sum_{j\geq 0} d_j(\beta)\,\chi_j(U), \qquad d_j(\beta) = \frac{2(j+1)}{\beta}\,I_{j+1}(\beta), \tag{8}$$

with I_k the modified Bessel function; $d_0(0) = 1$ and $d_j(0) = 0$ for $j \geq 1$.

3.3 Exact factorization

Proposition 1 (Exact solvability on the scalar graph)**.** *At every $n_{\max}$, with the plaquette set of Definition 1,*

$$Z(\beta) = \Big[e^{-\beta}\,\frac{2I_1(\beta)}{\beta}\Big]^{\beta_1}, \tag{9}$$

and every closed loop C in an l-block enclosing A unit squares has

$$\langle W_C\rangle \equiv \Big\langle \tfrac{1}{2}\mathrm{tr}\, U_C\Big\rangle = \left(\frac{I_2(\beta)}{I_1(\beta)}\right)^{A}. \tag{10}$$

Proof. The argument is the standard two-dimensional one [8, 7]. In each grid, fix the gauge $U_e = I$ on a spanning tree. The remaining β_1 links are in bijection with the unit squares (order the squares so that each adds one new non-tree edge), and the change of variables from those links to the plaquette holonomies is a product of left translations, which preserves Haar measure. The plaquette holonomies are therefore independent Haar variables, and Z is the product of β_1 one-plaquette integrals $e^{-\beta}d_0(\beta)$. A loop enclosing A squares has holonomy conjugate to a product of the A enclosed plaquette holonomies, each conjugated by a path holonomy; since each is distributed by a class function, Schur orthogonality gives $\langle\frac{1}{2}\mathrm{tr}\, U_C\rangle = \prod\langle\frac{1}{2}\mathrm{tr}\, U_P\rangle$, and the one-plaquette expectation is $d_1/(2d_0) = I_2/I_1$. □

The cycle-basis property and the one- and two-plaquette identities are tested in `tests/test_su2_wilson_gauge.py`, class `TestCertRunCoverageClosures` (the cycle-basis, single-plaquette and two-plaquette tests). At $n_{\max} = 3$, $\langle W_{P_A}\rangle = I_2/I_1 = 0.124,\ 0.240,\ 0.433,\ 0.719$ at $\beta = 0.5,\ 1,\ 2,\ 5$, and the six-edge loop around $P_A \cup P_B$ has $(I_2/I_1)^2$. The area law (10) holds at every β with string tension $-\ln(I_2/I_1)$ and no area–perimeter crossover; it is a consequence of factorization and carries no information about confinement. A confinement diagnostic with dynamical content needs plaquettes that are not independent, which the scalar graph does not supply.

3.4 Weak coupling: the curl form

Write $U_e = \exp(\frac{i}{2}A_e^a\sigma_a)$ with small real A_e^a. To second order the plaquette holonomy is $U_P = I + \frac{i}{2}\Phi_P^a\sigma_a - \frac{1}{8}\Phi_P^a\Phi_P^a I + \ldots$, where $\Phi_P^a = (B_2^\top A^a)_P$ is the circulation of A^a around P and the omitted terms are traceless at second order. Hence

$$1 - \tfrac{1}{2}\mathrm{Re}\,\mathrm{tr}\, U_P = \tfrac{1}{8}\sum_a (\Phi_P^a)^2 + O(A^3). \tag{11}$$

The $O(A^3)$ term does not vanish: the commutators $[A_e, A_{e'}]$ in $\log U_P$ feed a cubic vertex proportional to ϵ^{abc}.

Proposition 2 (Kinetic term)**.** *Around the trivial vacuum,*

$$S_W[U] = \frac{\beta}{8}\sum_{a=1}^{3}\big\langle A^a,\, B_2B_2^\top A^a\big\rangle + O(A^3). \tag{12}$$

The form $B_2B_2^\top$ is supported on the cycle space $\ker L_1$ *($L_1B_2 = 0$), and $L_1 + B_2B_2^\top = \Delta_1$ is the Hodge-1 Laplacian of the plaquette-filled complex, of full rank E on the scalar graph (Eq. (5) with* $\mathrm{rank}\, B_2 = \beta_1$*). The three components A^a decouple at quadratic order, each with coefficient* $1/8$ *per plaquette.*

At $n_{\max} = 3$, $\operatorname{rank} B_2 B_2^\top = 2 = \beta_1$, $\operatorname{rank} L_1 = 11 = E - \beta_1$, $L_1 B_2 = 0$, and $L_1 + B_2 B_2^\top$ has rank 13; the coefficient $1/8$ is checked against exact matrix-exponential holonomies on a triangle and on the full $n_{\max} = 3$ graph, with relative deviation below 5×10^{-3} at field size $\epsilon = 0.01$ (`tests/test_su2_wilson_gauge.py`, class `TestWeakCouplingKinetic`). The edge Laplacian L_1 is the exact (longitudinal) half of Δ_1, the half a gauge-fixing term supplies; the Wilson action supplies only the curl half. The curl form vanishes on bridges, so it gives no weight to the edges of the $l = 0$ chain.

Remark 1 (SU(N_c))**.** *With generators normalized to* $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ *and the action* $\beta \sum_P (1 - \frac{1}{N_c} \operatorname{Re} \operatorname{tr} U_P)$*, the per-plaquette, per-component coefficient is* $1/(4N_c)$*, the standard normalization of the Wilson action's naive continuum limit [3]. The* SU(3) *value* $1/12$ *is checked on the Bargmann–Segal graph of Paper 24 [22] (`tests/test_su3_wilson_s5.py`, class `TestWeakCouplingKinetic`).*

4 The abelian section

4.1 Maximal-torus reduction

Proposition 3 (Maximal-torus reduction)**.** *Restricting the links of (7) to the maximal torus,* $U_e = \operatorname{diag}(e^{i\theta_e}, e^{-i\theta_e})$*, gives the compact* $U(1)$ *Wilson action*

$$S_{U(1)}[\theta] = \beta \sum_P (1 - \cos \theta_P), \qquad \theta_P = (B_2^\top \theta)_P, \tag{13}$$

with gauge transformations $\theta \mapsto \theta + B^\top \chi$*.*

Proof. Products of diagonal matrices are diagonal, so $U_P = \operatorname{diag}(e^{i\theta_P}, e^{-i\theta_P})$ with reversed edges counted negatively, and $\frac{1}{2} \operatorname{Re} \operatorname{tr} U_P = \cos \theta_P$. □

The reduction is checked on a triangle plaquette and, with random diagonal links at $\beta = 2$, on the full $n_{\max} = 3$ graph, to machine precision (`tests/test_su2_wilson_gauge.py`, class `TestU1Reduction` and `test_u1_reduction_full_graph_n3_beta2`). On the torus the character restricts to $\chi_j = \sum_{k=0}^{j} e^{i(j-2k)\theta}$, and the expansion (8) becomes the Jacobi–Anger expansion $e^{\beta \cos\theta} = \sum_{n \in \mathbb{Z}} I_n(\beta) e^{in\theta}$. The reduction is a statement about the action, not about the path integral: Haar measure on SU(2) does not restrict to Haar measure on the torus, so compact $U(1)$ is a separate theory with the restricted action, not a sector of the SU(2) theory. It is solved on the scalar graph by the same argument: $Z = [e^{-\beta} I_0(\beta)]^{\beta_1}$ and $\langle \cos \theta_C \rangle = (I_1(\beta)/I_0(\beta))^A$. At weak coupling $S_{U(1)} = \frac{\beta}{2} \langle \theta, B_2 B_2^\top \theta \rangle + O(\theta^4)$: the abelian action is even in θ and has no cubic term. The same restriction for SU(3) on the Bargmann–Segal graph gives the coupled $U(1) \times U(1)$ character $\frac{1}{3}(\cos a_P + \cos b_P + \cos(a_P + b_P))$ (`tests/test_su3_wilson_s5.py`, class `TestCartanReduction`).

4.2 Dictionary

The ingredients of Wilson's construction correspond to graph-Hodge data on any oriented graph with a chosen plaquette set:

Wilson	Graph Hodge	Role
Site	vertex v	matter location
Link	oriented edge e	gauge-field location
Link variable	edge phase $e^{i\theta_e}$, $\theta \in C^1$	connection
Plaquette	4-cycle P, a column of B_2	face
Plaquette angle	$\theta_P = (B_2^\top \theta)_P$	curvature
Wilson loop	$\exp(i\langle c, \theta\rangle)$, $c \in \ker B$	gauge-invariant observable
Gauge transformation	$\theta \mapsto \theta + B^\top \chi$	coboundary shift
Kinetic operator	$B_2 B_2^\top$	curl form (Prop. 2)
Gauge-fixed propagator	$\Delta_1^{-1} = (B^\top B + B_2 B_2^\top)^{-1}$	Feynman-gauge photon
Scalar matter operator	$L_0 + m^2$, $L_0 = BB^\top$	covariant Laplacian at $\theta = 0$
Flat connections mod gauge	$\ker B_2^\top / \operatorname{im} B^\top$	H^1, Eq. (5)

The dictionary is standard and applies to every graph; it is recorded here because it fixes the notation for Section 5. On the scalar graph $H^1 = 0$, so every flat connection is pure gauge.

4.3 The $n_{\max} = 3$ spectra

The edge Laplacian of the scalar graph is an integer matrix, so its eigenvalues are algebraic. At $n_{\max} = 3$ its nonzero spectrum is the union of the nonzero Laplacian spectra of the three blocks: the path P_3 ($l = 0$, eigenvalues $1, 3$), the grid $P_2 \times P_3$ ($l = 1$, eigenvalues $1, 2, 3, 3, 5$), and the path P_5 ($l = 2$, eigenvalues $2 - 2\cos(k\pi/5)$, $k = 1, \ldots, 4$). With the two zero modes of the cycle space,

$$\operatorname{spec}(L_1) = \left\{0^{(2)}, \tfrac{3-\sqrt{5}}{2}, 1^{(2)}, \tfrac{5-\sqrt{5}}{2}, 2, \tfrac{3+\sqrt{5}}{2}, 3^{(3)}, \tfrac{5+\sqrt{5}}{2}, 5\right\}, \tag{14}$$

whose characteristic polynomial splits over $\mathbb{Q}(\sqrt{5})$. Collapsing each m-multiplet at fixed (n, l) to one vertex, with edge weights the number of crossing edges, gives a weighted graph on the six sectors (n, l) with Laplacian spectrum $\{0, 0, 0, 1, 3, 6\}$, characteristic polynomial $x^3(x-1)(x-3)(x-6)$; this is the base graph used in Paper 2 [20]. Both spectra are pinned exactly in `tests/test_paper25_s2_quotient.py`.

5 Compact $U(1)$ on the Rule-B graph

5.1 Setting and comparison class

On the Rule-B graph we take the compact $U(1)$ action (13) with $\mathcal{P}$ the 4-cycles. The plaquettes span the cycle space but are not independent ($b_2 > 0$), so the theory does not factorize, and the minimal closed 2-cycles of three plaquettes are where monopoles can be detected. Two features fix what this theory should be compared with. Each link lies in 8.8 plaquettes on average at $n_{\max} = 2$ and 37.5 at $n_{\max} = 3$, against $2(d-1)$ on $\mathbb{Z}^d$, and the mean degree grows with the cutoff; and the graph is irregular. Large plaquette multiplicity is the regime in which mean-field and large-dimension treatments of lattice gauge theory become controlled [8], and irregular lattices with plaquettes taken as the smallest loops are the setting of random-lattice gauge theory [9]. Those are the natural comparison classes. No diagnostic below was run against either, and none was run against a four-dimensional control; the diagnostics test compatibility with compact $U(1)$, which in three dimensions confines at every coupling through a dilute monopole plasma [10, 11] and in four dimensions has a Coulomb phase above $\beta_c \approx 1.01$ [12, 13].

The archived drivers and data for this section are in `tests/wilson_rule_b_support/`; every number quoted is recomputed or pinned by `tests/test_paper41_wilson_witnesses.py`.

5.2 Seven diagnostics

(1) Spectral dimension. From the heat-kernel return probability of L_0, fitted over the same fixed band $0.1 < P(t) - c/V < 0.5$ on both graphs, Rule B has d_s of about 2.1 to 2.3 for $n_{\max} = 3$–8, about 0.45 above the scalar graph (about 1.7 to 1.8 over the same band), with no trend in $n_{\max}$. A window that slides with graph size gives 1.86, 2.27, 2.54 at $n_{\max} = 3, 4, 5$ and passes 3 at $n_{\max} = 8$; that rise is an artifact of the window. The graph has growing degree and is not a lattice approaching a manifold.

(2) Migdal–Kadanoff flow. The bond-moving recursion [5, 6] $I_1(\beta')/I_0(\beta') = [I_1(\beta)/I_0(\beta)]^k$ with $k = \langle q_e \rangle$ flows to $\beta = 0$ with no nontrivial fixed point. This holds for any $k > 1$, because I_1/I_0 increases from 0 to 1, so it depends on the graph only through k. The recursion misses the four-dimensional transition on the hypercubic lattice as well, so the flow discriminates nothing.

(3) Gaussian Wilson loops. At weak coupling, $\langle W_C \rangle \simeq \exp(-\frac{1}{2\beta} \langle c, (B_2 B_2^\top)^+ c \rangle)$. Averaged over primitive closed non-backtracking walks of length $L = 4, 6, 8$, the exponent scales as L^α with $\alpha \in [0.89, 1.18]$ for $n_{\max} = 2$–5: a perimeter law, as for the free photon in three or four dimensions. This sector does not see compactness.

(4) Strong coupling. At leading order in small β, a loop bounding a surface of A plaquettes has $\langle W \rangle = (I_1/I_0)^A + \ldots$, with string tension $-\ln(I_1/I_0) \simeq \ln(2/\beta)$. This is a property of the expansion on any plaquette complex and says nothing outside the strong-coupling regime.

(5) Monopole density. With the DeGrand–Toussaint construction [13] on the three-plaquette closed 2-cycles (60 at $n_{\max} = 2$, 80 at $n_{\max} = 3$), each plaquette angle is split as $\theta_P = \bar{\theta}_P + 2\pi m_P$ with $\bar{\theta}_P \in (-\pi, \pi]$, and the integer charge on a 2-cycle S is $m_S = \sum_{P \in S} \pm m_P$. Metropolis Monte Carlo of (13) gives a density $\rho_M(\beta)$ that falls roughly exponentially on $0.05 \le \beta \le 1.0$; the fit $\rho_M = a\, e^{-c\beta}$ gives $c = 9.40$ and 8.95 at $n_{\max} = 2, 3$ ($R^2 = 0.981$ both). Above $\beta = 1$ the density is below the detection floor. Exponential suppression in β is generic to compact $U(1)$, and the measured range lies below the four-dimensional β_c, where the four-dimensional theory also confines; in four dimensions the monopole currents are closed loops that percolate in the confined phase and become small and sparse in the Coulomb phase [8, 13]. The data do not reach the coupling at which the two would differ.

(6) Full Monte Carlo Wilson loops. Loops are built as boundaries of clusters of A plaquettes. A fit of $\ln\langle W \rangle$ against A alone gives a positive, decreasing slope (0.377, 0.011, 0.003 at $\beta = 0.1$, 3, 10, $n_{\max} = 2$), but a joint fit $\ln W = -\sigma A - \mu L + \text{const}$ gives an area coefficient $|\sigma| < 0.05$ with $\mu > 5|\sigma|$ at every β at $n_{\max} = 2$: the loops are perimeter dominated and no area law is resolved. At $n_{\max} = 4$ the joint fit gives $\sigma = -0.080 \pm 0.002$ at $\beta = 0.3$, which we attribute to the cluster construction, in which the minimal perimeter is not monotone in the area. The exponential rate of the area-only slope, 1.196, matches that of the perimeter coefficient, 1.242, so the area-only fit measures the perimeter term; its disagreement with the dilute-gas prediction $c/2 = 4.70$ for a string tension [10] is open.

(7) Polyakov loops. On the product graph $G_B \times C_{N_t}$ at $n_{\max} = 2$, with spatial and temporal-rectangle plaquettes, the cohomology defect $E - V + 1 - \operatorname{rank} B_2$ of Eq. (5) is 1 for $N_t \ge 3$ and 0 for $N_t = 2$, where the temporal loop backtracks and $P \equiv 1$. The extra class is the temporal winding,

so the Polyakov loop $P(v)$ is a well-defined observable not generated by plaquettes. Shifting every temporal link of one time slice by φ leaves the action invariant and multiplies P by $e^{i\varphi}$, so $\langle P\rangle = 0$ at every β in finite volume and carries no phase information. The order parameter $\langle|P_{\rm lat}|\rangle$, with $P_{\rm lat} = V_B^{-1}\sum_v P(v)$, rises from the random-phase value 0.285 at $\beta = 0.3$ to 0.498 and 0.833 at $\beta = 1$, 3 ($N_t = 4$; 0.285, 0.341, 0.691 at $N_t = 8$). On a ten-site spatial graph this does not settle the phase, and it does not indicate persistent confinement.

5.3 Verdict

Compact $U(1)$ on the Rule-B graph is a well-defined, non-factorizing lattice gauge theory. Its diagnostics are compatible with compact $U(1)$: (2)–(4) hold for compact $U(1)$ on any plaquette complex, (1) gives a fixed-scale separation from the scalar graph rather than a dimension, (5) stops below the coupling where three and four dimensions differ, (6) resolves no area law, and (7) supplies a well-defined finite-temperature observable whose order parameter orders with β. No dimension is discriminated. The sizes are small ($n_{\max} \le 4$ for the Monte Carlo, $n_{\max} = 2$ for the product graph), and the comparisons the graph's high plaquette multiplicity calls for, against mean-field, large-dimension and random-lattice results, are not made.

6 Open questions

Matter coupling. The Rule-B vertices are Dirac labels, so covariant hopping $\bar\psi_v U_{vv'}\psi_{v'}$ on its dipole edges is the natural first matter-plus-gauge theory on a label graph; on the scalar graph matter coupling would only reproduce the factorized two-dimensional theories.

Comparison controls for Rule B. The diagnostics of Section 5 need, in order: mean-field and large-dimension predictions at the measured plaquette multiplicities; a four-dimensional hypercubic control run through the same code; monopole data above $\beta \approx 1$; and Wilson loops at fixed perimeter by construction.

Non-abelian links on Rule B. SU(2) links on the Rule-B graph give a non-factorizing non-abelian theory on a label graph; it has not been built.

7 Conclusion

On the scalar label graph the 4-cycles are the unit squares of planar grids and form a cycle basis, so SU(2) and $U(1)$ Wilson theories are the exactly solvable two-dimensional theories, $\langle W\rangle = (I_2/I_1)^A$ and $(I_1/I_0)^A$, at every $n_{\max}$; the weak-coupling kinetic term is the curl form $B_2B_2^\top$, which completes the edge Laplacian to the Hodge-1 Laplacian, with a nonzero cubic vertex in the non-abelian case. On the Dirac Rule-B graph the 4-cycles satisfy relations, compact $U(1)$ does not factorize, and seven standard diagnostics are compatible with compact $U(1)$ without discriminating a dimension; the graph's high plaquette multiplicity places it in the mean-field, large-dimension class, against which it has yet to be tested.

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